

CourtFlow - Report features reference

This document describes what each part of a match report means: report.json (machine-readable), the printable report.pdf, and how tiers group metrics.

Top-level fields (report.json)

- generated_at - UTC timestamp when the report was built.
- schema_version - Contract version for the JSON shape.
- status - e.g. computed when tracks were processed.
- tracking_mode - tracking (ByteTrack/BoT-SORT) or detection_only (no tracker).
- video - duration_seconds, fps, and related metadata from the source file.
- highlights - List of {start, end, reason} time windows (e.g. high_intensity from movement).

summary

- insight - One-line hero text (aligned with Tier 1 when tracks exist).
- tier_1_total_distance_m, tier_1_tracking_span_s - Tier 1 physical totals (meters, seconds).

players (per player id 1-4)

- active_duration_s - Time span from first to last sample.
- rank_by_distance, share_of_distance_pct - Rank and % of team total distance.
- avg_speed, max_speed - From movement metrics (normalized court scaling; see Tier 2 for km/h).
- sprint_count, acceleration_peaks, deceleration_count - Dynamics from segment speeds.
- lateral_movement_pct - Share of movement that is side-to-side.
- zone_coverage_pct - % of time in each of 6 court zones (0-5).
- net_pct, baseline_pct - % near net band vs baselines (spatial).

analytics

- heatmap_player_{1-4} - Per-player heatmap paths.
- team_spacing_m - team_1_avg_m (P1-P2), team_2_avg_m (P3-P4) mean separation in meters.

analytics_tiers.tier_1 (basic)

players: distance_m, active_duration_s, rank_by_distance, share_of_distance_pct, track_points.
insight - Same one-liner as summary when computed.

analytics_tiers.tier_2 (standard)

session: heatmaps paths, max_speed_m_s_any_player, sprint_count_all_players.
players: avg_speed_m_s / kmh, max_speed_m_s / kmh, sprint_count, distance_m.

analytics_tiers.tier_3 (detailed)

players: zone_coverage_pct (cells 0-5), net_pct, baseline_pct, lateral_movement_pct, acceleration_peaks, deceleration_count.

padel

wall_usage - wall vs ground bounce counts when bounce_events exist.
player_stats_sample - Sample rows of time-series stats when computed.

report.pdf (exported PDF)

Gener

Contains: header, summary, Tier 1-3 tables, embedded heatmap images (combined + per player), player table, padel stub, footer.

Heatmaps are read from the reports/ folder next to report.json.

API (optional)

GET /

GET /matches/{id}/report.pdf - Download PDF.